

Dataset

CFFM-Net

Ritesh Roshan

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01 Core / primary experimental stack

The entries follow the dataset taxonomy in the project dossier. “Core”, “Optional”, and similar labels describe the project plan.

RGBT-Tiny

Role: Primary benchmark

Project phase: Phases 1–3

115 paired visible–thermal sequences; ~93K frames; ~1.2M boxes; 7 classes; >81% of targets under 16×16; tracking IDs. Access is by request.

Access: <https://github.com/XinyiYing/RGBT-Tiny>

COCO 2017

Role: General detection / pretraining

Project phase: Phase 1

General object-detection benchmark and pretrained-weight source.

Access: <https://cocodataset.org/>

VisDrone2019-DET

Role: Aerial small-object detection

Project phase: Phase 1

Drone imagery for small-object detection and aerial robustness.

Access: <https://github.com/VisDrone/VisDrone-Dataset>

LLVIP

Role: Visible–infrared low-light detection

Project phase: Phase 1

Paired visible/infrared imagery; useful for RGB–thermal validation.

Access: <https://github.com/bupt-ai-cz/LLVIP>

M3FD

Role: RGB–thermal detection / fusion

Project phase: Phase 1

Visible/infrared multi-scenario detection benchmark.

Access: <https://github.com/JinyuanLiu-CV/TarDAL>

FLIR-aligned

Role: RGB–thermal validation

Project phase: Phase 1

Aligned visible/thermal ADAS data; access is through the provider form.

Access: <https://www.flir.com/oem/adas/adas-dataset-form/>

DroneVehicle

Role: Aerial RGB-infrared vehicle detection

Project phase: Phase 1

Paired aerial RGB/IR imagery with vehicle annotations.

Access: <https://github.com/VisDrone/DroneVehicle>

VisDrone-VID

Role: Video object detection

Project phase: Phase 2

Drone video detection benchmark.

Access: <https://github.com/VisDrone/VisDrone-Dataset>

MOT17

Role: Multi-object tracking

Project phase: Phase 2

Standard MOT benchmark for tracking evaluation.

Access: <https://motchallenge.net/data/MOT17/>

VT-VOD50

Role: RGB-thermal video detection

Project phase: Phase 2

50-video RGB-thermal benchmark listed in the dossier. No verified direct URL is supplied in the dossier.

Access: No verified direct URL supplied in the project dossier.

UVT-VOD2024

Role: Unaligned RGB-thermal video

Project phase: Phase 2

Unaligned video benchmark listed in the dossier for misalignment robustness. No verified direct URL is supplied in the dossier.

Access: No verified direct URL supplied in the project dossier.

MEVA

Role: Surveillance-style video stress test

Project phase: Phases 3-4

Ground/UAV video with thermal and consented actors; privacy-bounded use in the dossier.

Access: <https://mevadata.org/>

02 Supporting small / tiny-object datasets

The entries follow the dataset taxonomy in the project dossier. “Core”, “Optional”, and similar labels describe the project plan.

AI-TOD v2

Role: Tiny aerial objects

Project phase: Optional

28,036 images; 8 classes; mean object size ~12.8 px.

Access: <https://github.com/Chasel-Tsui/AI-TOD-v2>

AI-TOD

Role: Tiny aerial objects

Project phase: Optional

28,036 aerial images; 700,621 instances; mean object size ~12.8 px.

Access: <https://github.com/jwwangchn/AI-TOD>

SODA-D

Role: Small driving objects

Project phase: Optional

Driving small-object benchmark.

Access: <https://shaunyan22.github.io/SODA/>

SODA-A

Role: Small aerial objects

Project phase: Optional

Aerial small-object benchmark.

Access: <https://shaunyan22.github.io/SODA/>

TinyPerson

Role: Tiny pedestrian detection

Project phase: Optional

1,610 images and 72,651 persons.

Access: <https://github.com/ucas-vg/TinyPerson>

DOTA v1.5

Role: Aerial object detection

Project phase: Optional

2,806 images and 403,318 instances.

Access: <https://captain-whu.github.io/DOTA/>

xView

Role: Large-scale overhead imagery

Project phase: Supporting

Source dataset associated with AI-TOD; 1M+ objects across 60 classes.

Access: <https://xviewdataset.org/>

UAVDT

Role: Aerial video detection / tracking

Project phase: Optional

100 sequences and ~80K frames.

Access: <https://sites.google.com/view/grli-uavdt/>

SeaDronesSee

Role: Maritime search / rescue

Project phase: Optional

Maritime aerial imagery; CC0 according to the dossier.

Access: <https://github.com/Ben93kie/SeaDronesSee>

XS-VID

Role: Tiny-object video

Project phase: Optional

374 sequences and ~223K frames. No verified direct URL supplied in the dossier.

Access: No verified direct URL supplied in the project dossier.

TinySet-9M

Role: Multi-domain tiny objects

Project phase: Watch

April 2026 dataset noted in the dossier; details were marked unconfirmed.

Access: No verified direct URL supplied in the project dossier.

03 Supporting RGB / thermal / near-IR datasets

The entries follow the dataset taxonomy in the project dossier. “Core”, “Optional”, and similar labels describe the project plan.

KAIST

Role: RGB-thermal pedestrian detection

Project phase: Optional

Visible-thermal pedestrian benchmark; useful for cross-modality comparisons.

Access: <https://soonminhwang.github.io/rgbt-ped-detection/>

VT-VOD50

Role: RGB-thermal video detection

Project phase: Core for video

50 videos / paired RGB-thermal sequences listed in the dossier; no verified direct URL supplied there.

Access: No verified direct URL supplied in the project dossier.

UVT-VOD2024

Role: Unaligned RGB-thermal video

Project phase: Core for misalignment

~60K images and 11 classes listed in the dossier; no verified direct URL supplied there.

Access: No verified direct URL supplied in the project dossier.

VT-MOT

Role: RGB-thermal multi-object tracking

Project phase: Optional

582 sequence pairs; ~401K frames; ~3.99M boxes. No verified direct URL supplied in the dossier.

Access: No verified direct URL supplied in the project dossier.

VEDAI

Role: Aerial RGB + near-IR

Project phase: Optional

1,210 aerial images with RGB and near-IR imagery.

Access: <https://downloads.greyc.fr/vedai/>

CVC-14

Role: Far-IR + grayscale visible

Project phase: Cross-dataset

Poorly aligned visible/thermal pedestrian data; useful for cross-dataset robustness.

Access: <http://adas.cvc.uab.es/elektra/en/>

M3OT

Role: Multi-drone RGB-IR tracking

Project phase: Phase 4

2025 dataset; two drones and 10,790 RGB-IR pairs. No verified direct URL supplied in the dossier.

Access: No verified direct URL supplied in the project dossier.

MMOT

Role: Multispectral multi-object tracking

Project phase: Watch

2025 multispectral tracking dataset; eight bands noted in the dossier.

Access: No verified direct URL supplied in the project dossier.

DVT-VOD1000

Role: Unaligned RGB-thermal video

Project phase: Watch

July 2026 dataset; 1,000 unaligned videos and ~103,464 pairs, as listed in the dossier.

Access: No verified direct URL supplied in the project dossier.

DRGBT-1K

Role: RGB-thermal tracking with modality switching

Project phase: Watch

July 2026; 1,045 tracking sequences with modality switching, as listed in the dossier.

Access: No verified direct URL supplied in the project dossier.

04 Optional video / tracking datasets

The entries follow the dataset taxonomy in the project dossier. “Core”, “Optional”, and similar labels describe the project plan.

ImageNet VID

Role: Video object detection

Project phase: Optional

General video detection benchmark.

Access: <https://image-net.org/challenges/LSVRC/2015/>

MOT20

Role: Multi-object tracking

Project phase: Optional

Crowded-scene MOT benchmark.

Access: <https://motchallenge.net/data/MOT20/>

DanceTrack

Role: Multi-object tracking

Project phase: Optional

Tracking benchmark with appearance and motion variation.

Access: <https://dancetrack.github.io/>

SportsMOT

Role: Sports tracking

Project phase: Optional

Multi-object tracking in sports scenes.

Access: <https://github.com/MCG-NJU/SportsMOT>

BDD100K MOT

Role: Driving video tracking

Project phase: Optional

Large-scale autonomous-driving video benchmark.

Access: <https://www.vis.xyz/bdd100k/>

TAO

Role: Long-tail tracking

Project phase: Optional

Tracking Across Objects benchmark.

Access: <https://taodataset.org/>

LaSOT

Role: Single-object tracking

Project phase: Optional

Large-scale single-object tracking benchmark.

Access: <https://cis.temple.edu/lasot/>

05 General detection / pretraining datasets

The entries follow the dataset taxonomy in the project dossier. “Core”, “Optional”, and similar labels describe the project plan.

COCO-ReM

Role: General detection

Project phase: Optional

COCO-derived/revised benchmark listed in the dossier. No verified direct URL supplied.

Access: No verified direct URL supplied in the project dossier.

LVIS v1

Role: Long-tail instance detection

Project phase: Optional

Long-tail detection benchmark.

Access: <https://www.lvisdataset.org/>

Objects365

Role: Large-scale object detection

Project phase: Weights only

Large-scale pretraining source; the dossier recommends weights rather than full data because of budget.

Access: <https://www.objects365.org/>

Open Images V7

Role: General object detection

Project phase: Optional

Large-scale image dataset with detection annotations.

Access: <https://storage.googleapis.com/openimages/web/index.html>

PASCAL VOC

Role: General detection

Project phase: Debugging

Mature benchmark useful for pipeline debugging.

Access: <http://host.robots.ox.ac.uk/pascal/VOC/>

RF100-VL

Role: Roboflow detection datasets

Project phase: Optional

Collection used for transfer / validation in the dossier.

Access: <https://public.roboflow.com/>

06 Datasets to avoid or restrict

The project dossier explicitly identifies these datasets as withdrawn, restricted, avoided, or legacy-only.

DukeMTMC and derivatives

Status: Do not use

Restricted / withdrawn dataset policy in the dossier.

Brainwash

Status: Do not use

Restricted / withdrawn dataset policy in the dossier.

MS-Celeb-1M

Status: Do not use

Restricted / withdrawn dataset policy in the dossier.

Oxford Town Centre

Status: Avoid

Avoided by the dossier's dataset policy.

MSMT17

Status: Only if essential

Legacy comparison only; dossier flags imagery/publication concerns.

Market-1501 / CUHK03

Status: Legacy comparison only

Not part of the recommended core stack.

07 Recommended order of use

Start with RGBT-Tiny

Use it as the main paired visible-thermal benchmark for the core CFFM-Net experiments.

Establish visible-only baselines

Use COCO and VisDrone before introducing RGB-thermal fusion, so detector effects can be separated from fusion effects.

Validate cross-modality fusion

Use LLVIP, M3FD, FLIR-aligned, and DroneVehicle for complementary RGB-thermal experiments.

Move to temporal evaluation

Use sequence information from RGBT-Tiny together with VisDrone-VID and MOT17, followed by accessible RGB-thermal video datasets.

Stress tiny objects

Use AI-TOD v2, SODA-A/D, TinyPerson, DOTA v1.5, and other optional datasets when tiny-object diversity is needed.

Deployment / Phase 4

Use MEVA and Phase-4 datasets only when the live/video and privacy-bounded analytics components are ready.

08 Access notes

- Official or project-maintained pages are linked where a verified URL was identified. A project page is not necessarily a direct download link.
- RGBT-Tiny uses an access request process. Its official repository describes 115 paired sequences, 93K frames, 1.2M annotations, and tracking IDs.
- VisDrone provides separate DET, VID, SOT, MOT and crowd-counting resources from its official repository.
- LLVIP is described by its repository as available for academic and non-commercial purposes subject to its license terms.
- DroneVehicle's official repository provides dataset access information and preprocessing notes.
- Several newer or specialized datasets appear in the dossier without a verified direct URL. No invented links are included in this document.
- Check the current dataset license and terms before downloading, redistributing, publishing, or using a dataset commercially.